

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2025-26 (NEP)

DEPARTMENT OF MATHEMATICS**Paper Code: MATMJ-44 : Mechanics**

Time: 3 Hours

Maximum Marks: 60

Instructions: Attempt FIVE questions in total. All questions carry equal marks (12 marks each).

Question 1. State and prove the Principle of Virtual Work for a system of coplanar forces acting on a rigid body. [12]

Question 2. Derive the intrinsic equation of a Common Catenary $s = can\psi$, and obtain relation $y = c \cosh(x/c)$. [12]

Question 3. Derive the differential equation of a central orbit in pedal form $\frac{h^2}{p^3} \frac{dp}{dr} = F(r)$. [12]

Question 4. State Kepler's laws of planetary motion and deduce Newton's Law of Gravitation from Kepler's laws. [12]

Question 5. Derive tangential and normal accelerations of a particle moving along a plane curve. [12]